



**BINAY PowerLED Inspection Hand Lamps utilize a light source comprised of multiple high power white LEDs**

- Very low power consumption of 10 watts or less
- Long life (over 50,000 burning hours), covered under 3-year warranty
- Unbreakable shock-proof and vibration-resistant construction; cool light output with no heat radiation
- Glare-free light output, focusing only on the work area



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**POWERING LED TECHNOLOGIES WORLDWIDE SINCE 1983**

## BINAY Multichip LED Lamps

High reliability direct-retrofit LED bulbs for Control and Instrumentation applications



Ba9s bayonet pin cap



E-10 screw cap



Midget Flange cap



Midget Groove cap



5530 and 6844 slide base



Wedge base

Simple retrofits for existing filament bulb holders, in applications such as annunciator window (fascia) backlighting and illuminated pushbutton switches, where the reliability of indication is important or where lamp replacement is difficult or expensive

- Long life (typically 100,000 hours)
- Low power consumption;
- Low heat generation
- High brightness and wide-angle light emission
- Shock-proof and vibration resistant
- High reliability of indication
- One-time Expense



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## China's underwater great wall of robots

China is designing a manned deep-sea space station to hunt for mineral resources in South China Sea, according to a Bloomberg report. The said underwater wall essentially



A fleet of unmanned vehicles, namely sea drones, would be part of the underwater wall defence system (Image courtesy: www.seeker.com)

refers to a network of floating and submerged sensors designed to detect enemy submarines, involving underwater robots and drones.

The proposed location is significant because South China Sea is a highly-disputed area these days among China and its neighbours, especially Vietnam and The Philippines. The oceanic base would be built as deep as 3000 metres (9800-feet) below the surface, and would be inhabited by a full-time crew.

## Robots that can assemble telescopes in Space

Scientists, including one of Indian origin, have designed an extremely large telescope that can be assembled by a robot in Space to help enhance the ability of astronomers to peer deep into the cosmos.

Ground based telescopes are limited by atmospheric effects and by their fixed location on the Earth. Space based telescopes do not have those disadvantages but have other limits such as overall launch vehicle volume and mass capacity. Design of a modular Space telescope that overcomes restrictions on volume and mass may allow telescope components to be launched incrementally, enabling the design and deployment of extremely-large Space telescopes.

The robotically-assembled modular Space telescope (RAMST), designed by Nicolas Lee and colleagues at California Institute of Technology, USA, and NASA's Jet Propulsion Laboratory, USA, in the USA, focuses primarily on a robotic system that perform tasks in which astronaut fatigue would be a problem.